

Binary Arithmetic



BINARY ADDITION

- The steps used for a computer to complete addition are usually greater than a human, but their processing speed is far superior.

RULES

- $0 + 0 = 0$
- $0 + 1 = 1$
- $1 + 0 = 1$
- $1 + 1 = 0$ (With 1 to carry)
- $1 + 1 + 1 = 1$ (With 1 to carry)

BINARY ADDITION

EXAMPLE

$$\begin{array}{r} 1\ 0\ 0\ 1 \\ +\ 1\ 0\ 1\ 1 \\ \hline \end{array}$$

BINARY ADDITION

EXAMPLE

$$\begin{array}{r} 1001 \\ + 1011 \\ \hline \end{array}$$

0

BINARY ADDITION

EXAMPLE

$$\begin{array}{r} 1\ 0\ 1\ 0\ 1\ 1 \\ +\ 1\ 0\ 1\ 1 \\ \hline \end{array}$$

BINARY ADDITION

EXAMPLE

$$\begin{array}{r} 1\ 0\ 1\ 0\ 1\ 1 \\ +\ 1\ 0\ 1\ 1 \\ \hline 1\ 0\ 0 \end{array}$$

BINARY ADDITION

EXAMPLE

$$\begin{array}{r} 11 \ 01 \ 01 \ 1 \\ 1 \ 0 \ 1 \ 1 \\ \hline 0 \ 1 \ 0 \ 0 \end{array} +$$

BINARY ADDITION

EXAMPLE

$$\begin{array}{r} 11\ 01\ 01\ 1 \\ 1\ 0\ 1\ 1 \\ \hline 1\ 0\ 1\ 0\ 0 \end{array}$$

BINARY ADDITION

CHECK THE ANSWER

$$\begin{array}{r} 9 \\ 11 \\ \hline 20 \end{array} +$$



BINARY SUBTRACTION

- Computers have trouble performing subtractions so the following rule should be employed:

“ $X - X$ is the same as
 $X + -X$ ”

- This is where two's complement is used.

BINARY SUBTRACTION

RULES

2. Convert the number to binary.
3. Perform two's complement on the second number.
4. Add both numbers together.

BINARY SUBTRACTION

EXAMPLE 1

Convert $12 - 8$ using two's complement.

3. Convert to binary

$$12 = 00001100_2$$

$$8 = 00001000_2$$

● Perform one's complement on the 8_{10}

$$00001000_2$$

$$11110111_2$$

BINARY SUBTRACTION

EXAMPLE 1

2. Perform two's complement.

$$\begin{array}{r} 1111011_2 \\ 0000001_2^+ \\ \hline 1111100_2 \end{array}$$

6. Add the two numbers together.

$$= \cancel{1}00000100_2 \text{ (Ignore insignificant bits)}$$

BINARY SUBTRACTION

EXAMPLE 2

What happens if the first number is larger than the second?

Try $6_{10} - 10_{10}$

BINARY SUBTRACTION

EXAMPLE 2

2. Convert to binary

$$6 = 00000110_2$$

$$10 = 00001010_2$$

- Perform one's complement on the 10_{10}

$$00001010_2$$

$$11110101_2$$

BINARY SUBTRACTION

EXAMPLE 2

2. Perform two's complement.

1 1 1 1 0 1 0 1₂

0 0 0 0 0 0 0 1₂⁺

= 1 1 1 1 0 1 1 0₂

BINARY SUBTRACTION

EXAMPLE 2

2. Add the two numbers together.

0 0 0 0 0 1 1 0₂

1 1 1 1 0 1 1 0₂ ⁺

= **1** 1 1 1 1 1 0 0₂ (Ends with a negative bit)

BINARY SUBTRACTION

EXAMPLE 2

2. Perform one's complement on the result

1 1 1 1 1 1 0 0₂

0 0 0 0 0 0 1 1₂

5. Add 1 to the result.

0 0 0 0 0 0 1 1₂

0 0 0 0 0 0 0 1₂ ⁺

= 0 0 0 0 0 1 0 0₂

BINARY SUBTRACTION

EXAMPLE 2

2. We then add the sign bit back.

$$\begin{array}{l} 00000100_2 \\ = 10000100_2 \end{array}$$

BINARY MULTIPLICATION

- Multiplication follows the general principal of shift and add.
- The rules include:
 - $0 * 0 = 0$
 - $0 * 1 = 0$
 - $1 * 0 = 0$
 - $1 * 1 = 1$

BINARY MULTIPLICATION

EXAMPLE 1

Complete $15 * 5$ in binary.

3. Convert to binary

$$15 = 00001111_2$$

$$5 = 00000101_2$$

6. Ignore any insignificant zeros.

$$\underline{0000}1111_2$$

$$\underline{00000}101_2 \quad \times$$

BINARY MULTIPLICATION

EXAMPLE 1

2. Multiply the first number.

1 1 1 1₂

1 0 1₂ ×

1 1 1 1

$$1111 \times 1 = 1111$$

7. Now this is where the shift and takes place.

BINARY MULTIPLICATION

EXAMPLE 1

2. Shift one place to the left and multiple the second digit.

$$\begin{array}{r} 1111_2 \\ \times 101_2 \\ \hline 1111 \\ 00000 \end{array}$$

1111 x 0 = 0000

00000 ← Shift One Place

BINARY MULTIPLICATION

EXAMPLE 1

2. Shift one place to the left and multiple the third digit.

$$\begin{array}{r} 1111_2 \\ \times 101_2 \\ \hline 1111 \\ 0000 \\ 111100 \end{array}$$

$1111 \times 1 = 1111$

Shift One Place

BINARY MULTIPLICATION

EXAMPLE 1

2. Add the total of all the steps.

$$\begin{array}{r} 1111 \\ 00000 \\ 111100 \\ \hline 1001011 \end{array} \quad +$$

8. Convert back to decimal to check.

BINARY DIVISION

- Division in binary is similar to long division in decimal.
- It uses what is called a shift and subtract method.

BINARY DIVISION

EXAMPLE 1

Complete $575 / 25$ using long division.

2. $\begin{array}{r} 0 \\ 25 \overline{) 575} \end{array}$	• Take the first digit of 575 (5) and see if 25 will go into it. • If it can not put a zero above and take the next number.
2. $\begin{array}{r} 02 \\ 25 \overline{) 575} \end{array}$	• How many times does 25 go into 57? • TWICE

BINARY DIVISION

2.
$$\begin{array}{r} 02 \\ 25 \overline{) 575} \\ \underline{50} \\ 7 \end{array}$$

- How much is left over?
- $57 - (25 * 2) = 7$

2.
$$\begin{array}{r} 02 \\ 25 \overline{) 575} \\ \underline{50} \downarrow \\ 75 \end{array}$$

- Drop down the next value

BINARY DIVISION

2.
$$\begin{array}{r} 023 \\ 25 \overline{) 575} \\ \underline{50} \\ 75 \end{array}$$

- Divide 75 by 25
- Result = 3

2.
$$\begin{array}{r} 023 \\ 25 \overline{) 575} \\ \underline{50} \\ 75 \\ \underline{75} \\ 0 \end{array}$$

- Check for remainder
- $75 - (3 * 25) = 0$
- FINISH!

BINARY DIVISION

- Complete the following:
- 25/5

Step 1: Convert both numbers to binary.

$$25 = 1\ 1\ 0\ 0\ 1$$

$$5 = 1\ 0\ 1$$

Step 2: Place the numbers accordingly:

$$1\ 0\ 1 \overline{) 1\ 1\ 0\ 0\ 1}$$

BINARY DIVISION

Step 3: Determine if 1 0 1 (5) will fit into the first bit of dividend.

1 0 1 $\overline{) 1 1 0 0 1}$

1 0 1(5) will not fit into 1(1)

Step 4: Place a zero above the first bit and try the next bit.

BINARY DIVISION

Step 5: Determine if 1 0 1 (5) will fit into the next two bits of dividend.

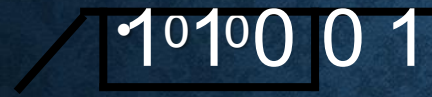
$$\begin{array}{r} 0 \\ 101 \overline{) 11001} \end{array}$$

1 0 1(5) will not fit into 1 1(3)

Step 6: Place a zero above the second bit and try the next bit.

BINARY DIVISION

- **Step 7:** Determine if 1 0 1 (5) will fit into the next three bits of dividend.

• 1 0 1 

- 1 0 1 (5) will fit into 1 1 0 (6)

- **Step 8:** Place a one above the third bit and times it by the divisor (1 0 1)

BINARY DIVISION

Step 9: The multiplication of the divisor should be placed under the THREE bits you have used.

$$\begin{array}{r} 001 \\ 101 \overline{) 11001} \\ \underline{101} \end{array}$$

A subtraction should take place, however you cannot subtract in binary. Therefore, the two's complement of the 2nd number must be found and the two numbers added together to get a result.

BINARY DIVISION

Step 10: The two's complement of 1 0 1 is 0 1 1

$$\begin{array}{r} 001 \\ 101 \overline{) 11001} \\ \underline{+ 011} \\ 001 \end{array}$$

BINARY DIVISION

Step 11: Determine if 1 0 1 will fit into the remainder 0 0 1. The answer is no so you must bring down the next number.

$$\begin{array}{r} 001 \\ 101 \overline{) 11001} \\ \underline{011} \\ 0010 \end{array}$$

BINARY DIVISION

Step 12: 1 01 does not fit into 0 0 1 0. Therefore, a zero is placed above the last bit. And the next number is used.

$$\begin{array}{r} 01 \\ 101 \overline{) 11001} \\ \underline{011} \\ 00101 \end{array}$$

The diagram illustrates the binary division process. The divisor is 101. The dividend is 11001. A box highlights the current step where 101 is subtracted from 110. Two downward arrows indicate the next steps: bringing down the next bit (0) to form 011, and then bringing down the final bit (1) to form 00101.

BINARY DIVISION

Step 13: 1 0 1 does fit into 1 0 1 so therefore, a one is placed above the final number and the process of shift and add must be continued.

$$\begin{array}{r} 00101 \\ 101 \overline{) 11001} \\ \underline{+ 011} \\ 00101 \\ \underline{+ 011} \\ 000 \end{array}$$

Binary Division



Step 14: The final answer is 1 0 1 (5) remainder zero.